BINF sights.

2026-T2

immunotherapy.

another future of Modern healthcare



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Freepik

We’re Back & Better than Ever

Immunotherapy is a branch of medicine that uses the body’s immune system to fight disease, most commonly cancer. Immunotherapy focuses on boosting or retaining immune cells so they can recognise and attack harmful targets with far greater precision

Over the past two decades, research has radically expanded what immunotherapy can do. Early treatments relied mainly on laboratory-made antibodies, but today research within this field is incredibly broad... it would be impossible to fit everything into this issue!

A huge part of this progress depends on working with biological data. In such a field it is not uncommon to analyse DNA and protein sequences, perhaps to predict how tumours evolve, or model antibody structures to understand how they bind their targets, maybe map how immune cells interact with cancer, and design new proteins that behave in specific ways inside the body. We hope to show you just some of the technologies/techniques powering immunotherapy outcomes.

Contents

Bacterial Pathogens in Cancer Immunotherapy 1

Cancer DESTROYED! 2

Can Generative AI help us to design new DNA sequences? 3

Bacterial Pathogens in Cancer Immunotherapy

Cancer has been shown to avoid immune destruction, and in some cases, hijack a patient's immune system to support its growth (Hanahan & Weinberg, 2026). Traditional cancer treatments, such as chemotherapy, are indiscriminate in their attacks and have long been known to weaken the immune system.

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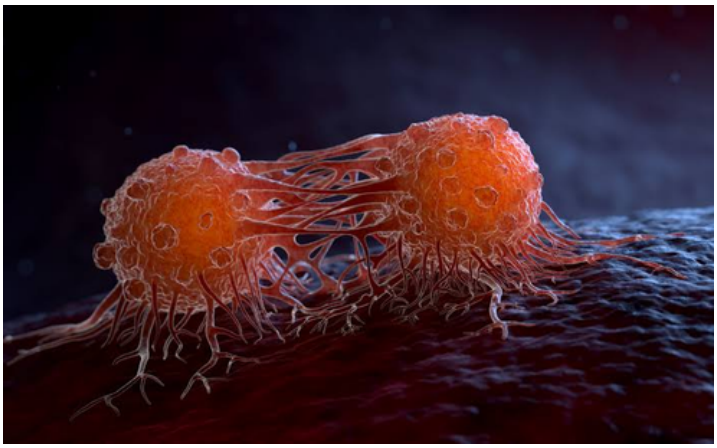


This weakens the body and increases vulnerability to infections, which can further complicate treatment (Sharma et al., 2023). Immunotherapy has been thought to avoid these downfalls, by working with the body's own immune system, rather than against, to fight the disease. It is worth noting that some of these treatment paths, such as checkpoint inhibitors (which enable the immune system to recognise and destroy cancer cells), may also cause increased infection risk, though generally less so than traditional approaches (Tangye et al., 2024).

A particularly innovative approach has been to exploit bacterial pathogens to fight cancer. A connection between the two was first made around 4,000 years ago, when local infection caused tumour eradication in some patients, though this was not taken advantage of until the 20th century (Kwon et al., 2024). Progress since then has been swift, with many developments in engineering bacteria for immunotherapy. Some of these have been proven and adopted as widely accepted treatments.

A first-line treatment for high-risk superficial bladder cancer is intravesical BCG, a live form of Mycobacterium bovis (closely related to human tuberculosis) delivered to the bladder via catheter. It has significantly higher odds of a complete response than intravesical chemotherapy (Sylvester et al., 2005).

Future developments for cancer immunotherapy are in the works, with focus now on new mechanisms, enhancing bacterial tumour colonisation, selectivity and safety, and extrapolation to other forms of the disease. The field is fast evolving, as are discoveries in synthetic biology, so information is developing very quickly. A recommended source of further reading for recent developments (at the time of writing this article) and past progress is Pico-Sánchez, E., & Fernández, L. Á. (2026). Progress in engineered bacterial cancer therapies. Knowledge is always changing, as are clinical approaches, so what is up-to-date now, might not be in future. That said, immunotherapy is already a very promising type of cancer treatment, and it will be exciting to see how it evolves.



Progress in engineered bacterial cancer therapies
Bacteria-based immunotherapy for cancer: a systematic review of preclinical studies
Bacillus Calmette-Guerin Versus Chemotherapy For The Intravesical Treatment Of Patients With Carcinoma In Situ Of The Bladder
Intravesical Immunotherapy (BCG) for Bladder CancerSharma, A., Jasrotia, S., & Kumar, A. (2023). Effects of Chemotherapy on the Immune System: Implications for Cancer Treatment and Patient Outcomes
Immunotherapy
Hallmarks of Cancer: The Next Generation
Exploiting bacteria for cancer immunotherapy

Cancer DESTROYED!

Immunotherapy has transformed cancer treatment by using the patient's own immune system to recognise and eliminate tumour cells, offering an alternative to conventional therapies such as chemotherapy and radiation. Personalised mRNA cancer vaccines represent a notable development within this field, with each vaccine designed around the specific profile of an individual patient’s tumour.

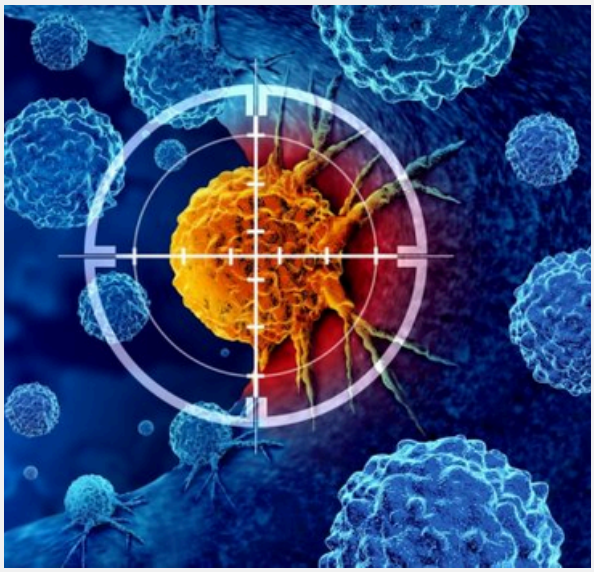
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In contrast to more traditional vaccines, which serve as preventative action against infectious diseases, personalised mRNA cancer vaccines are designed to elicit an immune response against cancer already present in the body.

The main targets of these vaccines are neoantigens, abnormal proteins resulting from mutations in the cancerous tumour cells. Because neoantigens are absent from healthy tissues, the immune system can readily recognise them as foreign and mount targeted responses to cells which express them. This separates neoantigens from other tumour-associated antigens, which are also found in normal tissues and are frequently subject to immune tolerance. As a result, neoantigens are better-suited targets for mRNA vaccines, allowing a stronger, more specific immune response while minimising damage to healthy cells.

Constructing a personalised vaccine begins with sequencing DNA and RNA from a patient's tumour and creating a profile of mutations unique to that individual's cancer.

Researchers then analyse that profile to predict which tumour-specific mutations are most likely to produce neoantigens capable of eliciting a strong immune response. These neoantigens are subsequently encoded into an mRNA vaccine, which, after being administered, causes the host’s cells to take up the mRNA and translate it into the corresponding neoantigen proteins. These neoantigen proteins are then presented on the surface of the cell by the major histocompatibility complex (MHC) molecules, activating tumour-specific CD4+ and CD8+ T cells. This enables the immune system to recognise and eliminate cancer cells expressing the relevant neoantigens.



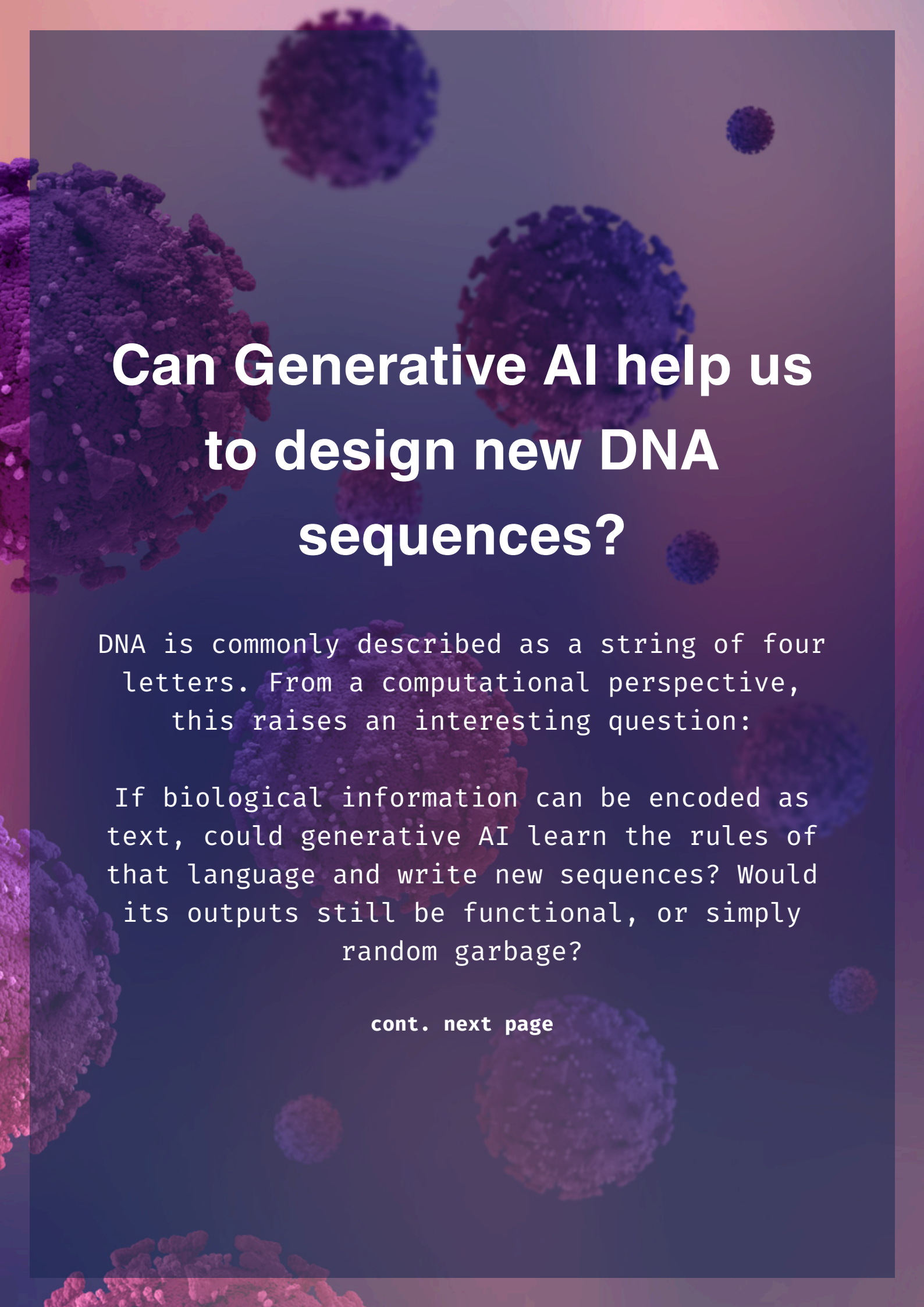
Early clinical studies provided initial evidence of the therapeutic potential of this approach. Sahin et al. demonstrated that personalised RNA vaccines elicited broad immune responses in patients with advanced melanoma. At the same time, Ott et al. reported that personalised neoantigen vaccination was safe, feasible, and capable of inducing durable T-cell immunity. These findings provided the foundation for the Phase IIb KEYNOTE-942 trial, in which the personalised mRNA vaccine mRNA-4157 (V940), administered alongside pembrolizumab, significantly reduced the risk of recurrence or death compared with pembrolizumab alone in patients with resected high-risk melanoma. This trial constitutes the strongest clinical evidence to date supporting personalised neoantigen vaccination.

Several challenges remain before personalised mRNA cancer vaccines can be adopted as routine clinical treatment.

A substantial proportion of tumour mutations fail to produce effective neoantigens, tumour cells can evade immune detection through loss of neoantigen expression or reduced antigen presentation, and manufacturing an individualised vaccine for each patient remains costly and time-intensive. Recent reviews indicate that continued progress in neoantigen identification, mRNA vaccine design, and clinical implementation is likely to extend this approach to a wider range of malignancies, consolidating its position as one of the most promising directions in precision cancer immunotherapy.



Individualised neoantigen therapy mRNA-4157 (V940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (KEYNOTE-942) - The Lancet
An Immunogenic Personal Neoantigen Vaccine for Melanoma Patients - PMC
Personalized RNA mutanome vaccines mobilize poly-specific therapeutic immunity against cancer - Nature
Neoantigens: promising targets for cancer therapy - Nature
Cancer vaccines: advances, hurdles, and future directions - PMC



Can Generative AI help us to design new DNA sequences?

DNA is commonly described as a string of four letters. From a computational perspective, this raises an interesting question:

If biological information can be encoded as text, could generative AI learn the rules of that language and write new sequences? Would its outputs still be functional, or simply random garbage?

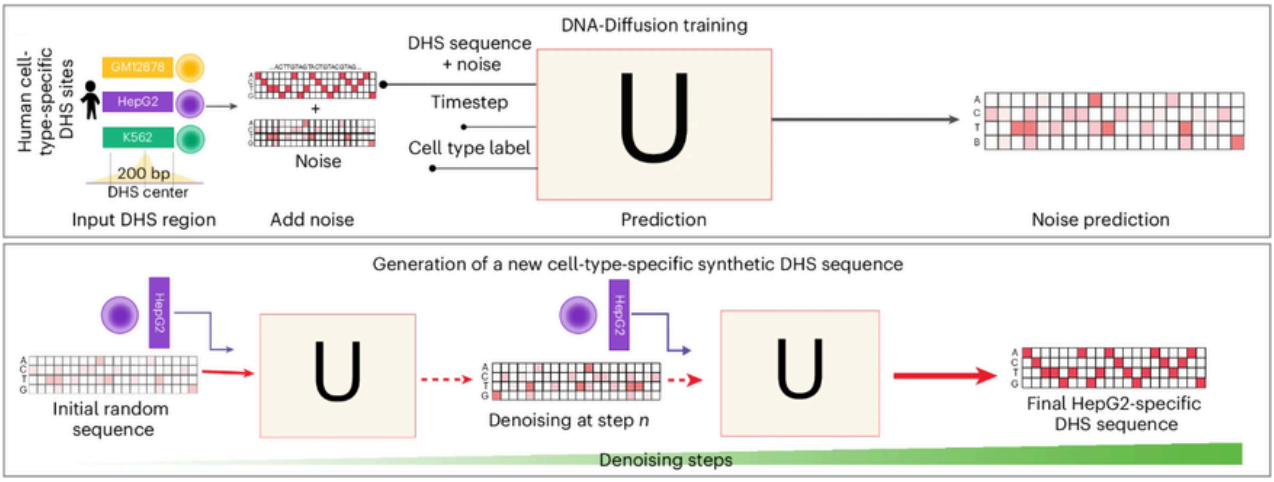
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DNA Diffusion is a generative model designed to produce cell-type specific regulatory sequences. Testing of its sequences in live cells showed that they outperformed regular regulatory elements, significantly enhancing chromatin accessibility and gene expression. This presents an exciting opportunity for future gene editing experiments, magnifying our current control of the genome.

With a name inspired by the image generation framework ‘Stable Diffusion’, DNA Diffusion is an example of a diffusion model. Diffusion models ‘learn’ by adding noise to an input until it becomes unrecognisable. In this case, the input is a DNA sequence from a specific cell type, and the noise may be random mutations in the sequence.

Each noise-adding step reinforces the model’s understanding of how the original and noisy sequence are related through a series of mathematical functions and weightings.

Generation is performed through a U-Net architecture. This involves a series of downsampling and upsampling steps. Downsampling is performed by compressing information to encode high level features, shrinking the input size. This can be visualised as encoding several bases into a single motif. Upsampling steps involve applying the model’s learned knowledge to the noisy downsampled precursor to extract encoded information into a more precise sequence. The final output is a 200 bp enhancer.



The ability to generate new sequences is meaningless if they simply copy pre-existing elements in the cell. An analysis performed using the BLAST-like alignment tool (BLAT) showed that less than 5% of generated sequences had a significant alignment to the training sequences. In comparison, large alignments occured 23.1% of the time in randomly selected sequences, indicating that the generated sequences are highly unique.

To demonstrate the potential uses of DNA Diffusion in healthcare, the authors generated a new enhancer for Axin-related Protein 2 (AXIN2). AXIN2 is often silenced in leukemic B cells and is associated with a worsened prognosis. Interestingly, insertion of several generated enhancers led to a strong reactivation of AXIN2 in leukemic cells. Furthermore, when enhancers were introduced into cell lines that they were not designed for, AXIN2 remained silenced. Thus, DNA Diffusion has the potential to be used in future gene therapies without off-target effects being a large concern.



Overall, DNA Diffusion provides the ability to quickly generate candidate enhancer sequences. The fact that these can be validated computationally with strong association with in vivo effects provides an opportunity to accelerate the design and evaluation cycles in gene regulation studies and gene therapies.

Cell-type-directed design of synthetic enhancers - PMC

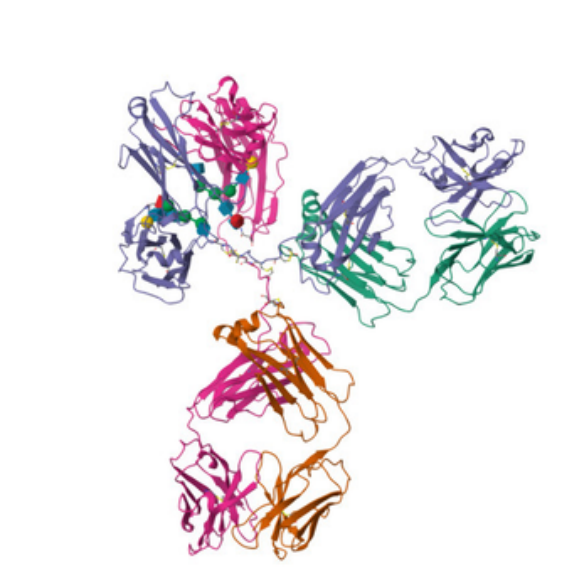
Designing synthetic regulatory elements using the generative AI framework DNA-Diffusion | Nature Genetics

Eukaryotic enhancers: common features, regulation, and participation in diseases - PMC

PROTEIN OF THE ISSUE

IgG

IMMUNOGLOBULIN G



IgG is the most abundant antibody in human serum, being ~75% of circulating immunoglobulins.
It has the molecule with the distinctive Y-shaped structure and composed of 4 subclasses.

At a molecular level, it is made of:
- Two heavy chains (γ-chains)
- Two light chains (κ or λ)
- A variable region that determines antigen specificity
- A constant region (Fc) that recruits immune effector functions

One notable property is the hinge region, a short, flexible segment which connects the antigen-binding arms (Fab) to the signalling tail (Fc). The hinge acts like a molecular joint, letting the Fab arms move, spread, and angle themselves to grab targets more effectively. More flexibility means better reach, but also makes the antibody easier to break down, so different IgG subclasses use different hinge lengths to balance strength and stability.

This flexibility also makes the hinge hard to predict with computer models, since it doesn't hold a fixed shape; as such, researchers use molecular dynamics simulations (computer methods that model movement over time) to understand how the hinge behaves.

These simulations matter because the Fc region, which triggers immune responses, can be engineered to change how strongly IgG activates the immune system. (A computation tool to design safer and effective immunotherapies!). For further interest, studies (Rouge et al., Frontiers in Immunology) have shown hinge flexibility affecting receptor binding.

Contact us



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Kind regards,

The BINFSOC Team

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